

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [125, 123, 116, 127, 128, 137, 120, 127, 114, 143]

Tap amplitudes (pixels): [145.64683312726027, 149.4958193395387, 140.06427096158393, 140.34600101178515, 138.3979768638256, 136.24243098242192, 144.03124660989366, 136.60527076214885, 138.80922159568507, 136.40014662748717, 138.45215780189199]

GPT-5

Analysis of extracted features for MDS-UPDRS 3.4 (Finger Tapping)

- Task: 10 rapid, large taps; evaluate speed, rhythm interruptions/hesitations, halts/freezes, and amplitude decrement over the 10 taps.
- Provided data:
 - Number of taps reported: 11 (suggests 10 intervals between taps for period list; amplitudes list has 11 values—one per tap)
 - Periods (frames) for 10 intervals: [125, 123, 116, 127, 128, 137, 120, 127, 114, 143]
 - Amplitudes (pixels) for 11 taps: [145.6, 149.5, 140.1, 140.3, 138.4, 136.2, 144.0, 136.6, 138.8, 136.4, 138.5] (rounded)

Rhythm interruptions / hesitations

- Interruptions are typically inferred from pronounced variability or outliers in periods, or explicit halts/freezes (not present as explicit flags here).
- Period variability: values range 114–143 frames. The sequence is relatively stable with mild variability; one longer interval at the end (143) and a slightly longer one at 137. No clear evidence of >5 interruptions or a longer arrest. This pattern aligns better with slight-to-mild slowing/variability rather than frequent interruptions.

Speed

- Without absolute calibration (frames-to-seconds), we compare relative consistency. Periods are mostly clustered (114–137) with one longer at 143. This suggests at most slight to mild slowing rather than moderate/severe.

Amplitude decrement

- Compare early vs later taps:
 - Early taps (1–3): ~145.6, 149.5, 140.1
 - Mid taps (4–7): ~140.3, 138.4, 136.2, 144.0
 - Late taps (8–11): ~136.6, 138.8, 136.4, 138.5
- There is a decrease from the peak at tap 2 (149.5) to most subsequent taps (~136–140), with occasional partial recoveries (tap 7 ~144.0). The decrement is present by mid-sequence (taps 4–6 are down to ~136–140) and then remains in that range. This is more than a “near the end” drop; it appears by midway and persists, which matches the criterion for score 2.c (amplitude decrements midway through the 10-tap sequence).
- It does not start immediately after the first tap (tap 2 actually increases), so it does not meet the 3.c threshold (decrement starting after the 1st tap).

Overall mapping to criteria

- Interruptions: none clearly excessive; no freeze → not 3.a or 4.
- Speed: likely slight to mild slowing → compatible with 1.b or 2.b, but amplitude criterion is clearer.
- Amplitude: decrement appears by mid-sequence → matches 2.c.

Final judgment: The clearest qualifying criterion is “amplitude decrements midway through the 10-tap sequence,” which corresponds to Mild (2).

Score: 2
